SBC Vacuum Controller — Shopping List

Raspberry Pi 4 Model B or Raspberry Pi 5 + RPi5VacuumIO rev A.1 board, for the FEI DB235 vacuum system. One controller. Prices in USD, checked at PiShop.us on 15 Oct 2026 unless marked *est.* (typical street price, not checked against a listing). Branch `Operations @ 15f6a0a`.

Summary

	Variant A: Pi 4 Model B	Variant B: Pi 5
Computer + storage	$186–221	$292–309
Board side, safety chain (common)	$209–388 + board quote	$209–388 + board quote
Sockets, connectors and cables (common)	$675–1,055	$675–1,055
Total, excluding board fabrication	**$1,070–1,664**	**$1,176–1,752**
Boot device	USB 3 SSD (BOOT_ORDER 0xf14)	NVMe on M.2 HAT+ (BOOT_ORDER 0xf416)
Clock after a power cut	Wrong until NTP is back (no RTC)	Correct (RTC + battery)
OS-image test before hardware	QEMU raspi4b (QEMU 9.0+)	None (no QEMU Pi 5 machine)
Controller-logic test	Python emulator, model 4B	Python emulator, model 5
config.txt for the board	i2c_arm, enable_uart, disable-bt, uart5	i2c_arm, enable_uart, uart4-pi5
Profile settings	PiModel 4B; RS232 node resolved on target	PiModel 5; RS232 /dev/ttyAMA4

Which one. Both work with the same board, software and wiring; nothing in the PCB or the code is tied to one model. The repository defaults to the Pi 4B (`PiModel: 4B`) so the OS image can be checked in QEMU. For a controller that runs 24/7, the Pi 5 is the stronger choice: the RTC keeps alert and log timestamps right after a power cut, and NVMe is more robust than a USB-attached SSD. It costs about $90–105 more. A sensible split is a Pi 4B on the bench for QEMU and emulator work and a Pi 5 in the tool. The install package (`sbc-deploy`) detects the model and writes the matching `config.txt` and profile.

About a quarter of the total is the safety chain (E-stop + safety relay). It is required on the tool: the board only grants permissives and must never be the only thing between the operator and hazardous energy.

Variant A — Raspberry Pi 4 Model B

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
Raspberry Pi 4 Model B, 2 GB (SC0193)	1	$67.50	$67.50	PiShop.us, in stock	Matches QEMU *raspi4b* (2 GiB) for OS-image tests. The SBC service uses well under 300 MB. 4 GB is $100, 8 GB $165.
Raspberry Pi 15 W USB-C PSU (5.1 V / 3 A)	1	$8.80	$8.80	PiShop.us	The Pi 4B's rated supply. USB devices share 1.2 A.
Case with fan, or DIN-rail Pi 4 enclosure + heatsink	1	$10–25	$10–25	est.	Cabinets run warm; throttling slows boot and logging.
Boot disk: USB 3 SSD, e.g. Samsung T7 500 GB	1	$60–80	$60–80	est.	Blue USB 3 port. BOOT_ORDER=0xf14 (USB, then SD). Alternative: USB-SATA UASP adapter + 2.5" SSD.
microSD, Raspberry Pi A2 64 GB (rescue / first boot)	1	$39.95	$39.95	PiShop.us (Sep 2026)	First boot, EEPROM update, rescue image. Keep it in the slot.
Subtotal			**$186–221**		

Variant B — Raspberry Pi 5

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
Raspberry Pi 5, 4 GB (SC1111)	1	$110.00	$110.00	PiShop.us, in stock	The 2 GB ($77.50) is out of stock at PiShop; 8 GB is $175. No QEMU machine for the Pi 5.
Raspberry Pi 27 W USB-C PSU (5.1 V / 5 A)	1	$12.95	$12.95	PiShop.us	Needed for the full 1.6 A USB budget. With a 3 A supply the Pi 5 limits USB to 600 mA.
Raspberry Pi Active Cooler	1	$5–6	$5–6	est. (list $5)	Fits under the M.2 HAT+.
Raspberry Pi M.2 HAT+ SSD kit, 256 GB	1	$99.00	$99.00	PiShop.us	HAT+, official 256 GB NVMe, 16 mm stacking header. BOOT_ORDER=0xf416 (NVMe, USB, SD). Cheaper: M.2 HAT+ alone $12 + your own 2230/2242 NVMe.
RTC battery, Panasonic ML-2020 with JST lead (official)	1	$5–6	$5–6	est.	**Pi 5 only.** Keeps the clock through a power cut, so vacuum logs and alerts have correct timestamps before NTP is back.
Case for Pi 5 + HAT (e.g. EDATEC ED-PI5CASE-EXT), or DIN-rail carrier	1	$20–35	$20–35	est.	Must clear the stacking header; the board's ribbon still goes to the GPIO header (see note).
microSD, Raspberry Pi A2 64 GB (rescue / first boot)	1	$39.95	$39.95	PiShop.us (Sep 2026)	First boot, EEPROM update, rescue image.
Subtotal			**$292–309**		

Pi 5 stacking note. The M.2 HAT+ sits on the 40-pin header through the 16 mm stacking header; plug the board's 40-way ribbon onto the pins that pass through it (pin 1 aligned). The HAT+ uses only PCIe and the ID EEPROM lines (GPIO0/1), which the I/O board does not use.

Common to both variants — board, power, safety chain

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
RPi5VacuumIO rev A.1 board, fabricated and assembled	1	quote	quote	board-house quote	Gerbers, BOM (62 lines) and pick-and-place in `RPi5VacuumIO/fab/`. The same board serves Pi 4B and Pi 5.
24 V DIN-rail PSU, 60–100 W (e.g. Mean Well HDR-60-24)	1	$20–30	$20–30	est.	Feeds J1. Separate from the Pi's USB-C supply.
E-stop (mushroom, 2 NC) **+ safety relay** (Pilz PNOZ s / Phoenix PSR)	1	$150–300	$150–300	est.	Not optional on the tool. One output contact to J2, one into the tool's own enable chain.
DIN-rail PCB carrier or M3 standoffs for the board	1	$10–20	$10–20	est.	
500 Ω 0.1 % resistors (only for 4–20 mA gauge outputs)	4	$1–2	$4–8	est.	Across AI–0V, gives 2–10 V.
Commissioning: USB microSD reader, micro-HDMI cable, USB-TTL 3.3 V cable	1	$25–30	$25–30	est.	With disable-bt the console UART is the RS-485 port: use SSH or HDMI.
Subtotal			**$209–388**		

Sockets, connectors and cables

Everything that plugs into the board or runs from the cabinet to the tool. Phoenix plug prices are Mouser single-piece prices and the Belden 8761 price is PacRad's per-foot price, both checked 17 Oct 2026; the rest are estimates. Cable quantities are in feet, for a cabinet about 3 m (10 ft) from the pump controllers and gauges. Subtotal for this section: **$675–1,055**.

BOM correction. Three Phoenix numbers in the board BOM were wrong and are corrected in the regenerated fab package: J31–J34 plug 1840395 (a 5-position part) → 1840405; J41/J42 plug 1840411 → 1840421; J41/J42 header 1844265 (7-position) → 1844278. The PCB footprints were already correct.

Board mating plugs (Phoenix Contact) — not in an assembly quote, order separately

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
1757019 MSTB 2,5/2-ST-5,08 — J1 24 V in	1	$1.81	$1.81	Mouser, qty 1, 17 Oct 2026	
1757022 MSTB 2,5/3-ST-5,08 — J11–J18 relay contacts	9	$2.71	$24.39	Mouser, qty 1, 17 Oct 2026	8 + 1 spare.
1840366 MC 1,5/2-ST-3,5 — J2 E-stop loop	1	$2.34	$2.34	Mouser, qty 1, 17 Oct 2026	
1840379 MC 1,5/3-ST-3,5 — J4 RS-485, J5 RS-232	2	$3.52	$7.04	Mouser, qty 1, 17 Oct 2026	
1840382 MC 1,5/4-ST-3,5 — J21–J24 solenoids	4	$4.62	$18.48	Mouser, qty 1, 17 Oct 2026	
1840405 MC 1,5/6-ST-3,5 — J31–J34 digital inputs	5	$7.03	$35.15	Mouser, qty 1, 17 Oct 2026	4 + 1 spare. **Corrected:** the BOM said 1840395, a 5-position plug.
1840421 MC 1,5/8-ST-3,5 — J41, J42 analog inputs	2	$9.32	$18.64	Mouser, qty 1, 17 Oct 2026	**Corrected:** the BOM said 1840411.
Subtotal			**$107.85**		

Raspberry Pi link and network

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
40-way IDC ribbon, female–female, 150 mm (2×20)	1	$3–5	$3–5	est. (The Pi Hut £3)	Pi header → J3, pin 1 to pin 1 (red stripe).
Cat6 patch cable, 2 m	1	$5–8	$5–8	est.	Pi → control network.
RJ45 panel / DIN-rail feed-through coupler (optional)	1	$10–15	$10–15	est.	Network into the cabinet without opening it.
Subtotal			**$18–28**		

Field cables (≈ 3 m / 10 ft per run — scale to the real distances)

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
Belden 8761, 1 pair 22 AWG, foil shield — gauges AI1–AI4, pump speed AI5–AI7	70	$1.70	$119.00	PacRad, per ft, 17 Oct 2026	7 runs × 10 ft (qty in ft). Shield to PE at the cabinet end only.
Belden 3105A, 1 pair 22 AWG, 120 Ω — RS-485 daisy chain (J4)	30	$3.20–4.00	$96–120	est. (spool price $3.16/ft, Markertek)	Board → turbo → UH1 → UH2, 3 segments × 10 ft. No stubs over 0.3 m.

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
Control cable 6 × 22 AWG, shielded (e.g. LiYCY 6×0.34) — pump start, ready, fault, common	50	$1–2	$50–100	est.	One per pump (4) + HV permissive: 5 runs × 10 ft.
Solenoid cable 2 × 18 AWG (0.75 mm²) — V1–V8	80	$0.40–0.80	$32–64	est.	8 runs × 10 ft.
Sensor cable 3 × 22 AWG — valve switches, door, air, N2, water (DI9–DI16)	80	$0.40–0.80	$32–64	est.	8 runs × 10 ft.
Hook-up wire 18 AWG stranded, red / black / blue, 25 ft each	1	$20–30	$20–30	est.	24 V supply (J1), E-stop loop (J2), safety relay.
Subtotal			**$349–497**		

Device-side connectors

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
Mating plugs for the pump controllers and gauges (D-sub, RJ45/FCC68 or M12, per manual)	10	$5–15	$50–150	est.	Confirm each type and pinout against the controller and gauge manuals (WIRING.md).
DE-9 female solder plug + backshell — J5 RS-232 (optional)	1	$4–8	$4–8	est.	Only with an RS-232 gauge controller.
Subtotal			**$54–158**		

Cabinet wiring hardware

Item	Qty	Unit (USD)	Line (USD)	Price source	Notes
Feed-through terminal blocks, 2.5 mm² (Phoenix PT 2,5 or WAGO 2002-1201)	40	$0.80–1.50	$32–60	est.	Tool cables land here, not directly on the board plugs.
End plates, end stops, jumpers and markers for the terminal blocks	1	$10–20	$10–20	est.	
DIN rail 35 mm, 1 m	1	$5–10	$5–10	est.	
Wire duct 25 × 40 mm, 2 m	1	$15–25	$15–25	est.	
Cable glands M16 / M20, IP68	12	$1–2	$12–24	est.	One per cable entering the cabinet.
EMC shield clamps + earth bar	1	$25–50	$25–50	est.	For the 8761, 3105A and control-cable shields.
Insulated ferrule assortment 0.25–1.5 mm²	1	$12–20	$12–20	est.	
Ferrule crimp tool (square, 0.25–6 mm²)	1	$25–40	$25–40	est.	
Wire markers / labels	1	$10–15	$10–15	est.	
Subtotal			**$146–264**		

Board-side sockets (fitted by the assembler — part of the board quote, listed for reference)

Board terminals	Phoenix part	Type	Qty
J1	1757242	MSTBA 2,5/2-G-5,08	1
J11–J18	1757255	MSTBA 2,5/3-G-5,08	8
J2	1844210	MC 1,5/2-G-3,5	1
J4, J5	1844223	MC 1,5/3-G-3,5	2
J21–J24	1844236	MC 1,5/4-G-3,5	4
J31–J34	1844252	MC 1,5/6-G-3,5	4
J41, J42	1844278	MC 1,5/8-G-3,5 (corrected from 1844265, a 7-position header)	2
J3	–	2×20 2.54 mm shrouded box header (Pi ribbon)	1

Tool cables should land on the DIN-rail terminal blocks first and reach the board plugs through short cabinet jumpers, so a board swap does not mean re-terminating field cables. Shields: one end only, at the cabinet (WIRING.md step 9).

Wired connection map (summary)

The deploy package ships this map as machine-readable `wiring-map.json` and checks it on the Pi before the service is enabled (pin functions, I2C addresses, serial nodes). Field-side terminal wiring is in `RPi5VacuumIO/WIRING.md`.

Raspberry Pi 40-pin header → board J3 (identical on Pi 4B and Pi 5)

Pin	BCM	Net	Function on the board	Pi setting
1	3V3	3V3	Board logic supply (also pin 17)	
2	5V	–	Deliberately not connected on J3 (also pin 4)	
3	GPIO2	SDA1	I2C1 → U1 0x20, U3 0x21, U9 0x48, U10 0x49	dtparam=i2c_arm=on
5	GPIO3	SCL1	I2C1 clock	
6	GND	GND	Also 9, 14, 20, 25, 30, 34, 39	
8	GPIO14	TXD0	RS-485 (MAX3485) → J4	Pi 4B: dtoverlay=disable-bt
10	GPIO15	RXD0	RS-485 receive	
12	GPIO18	RS485_DE	Driver enable, R14 pull-down	output
13	GPIO27	DI_INT	U3 INTA, open drain, pulled up	input
15	GPIO22	IO_RESET_N	Expander reset; R10 pull-down holds outputs off	output
16	GPIO23	ADC_ALRT	ADS1115 ALERT/RDY, wired-AND	input
29	GPIO5	ESTOP_OK_N	Low = E-stop rail live (opto)	input
31	GPIO6	SAFE_OK_N	Low = output (watchdog) rail live	input
32	GPIO12	232TX	RS-232 (MAX3232) → J5	4B: uart5 · 5: uart4-pi5
33	GPIO13	232RX	RS-232 receive	
35	GPIO19	HEARTBEAT	1 kHz into the K10 charge pump	output (PWM)
37	GPIO26	RUN_LED	Blue RUN LED D11	output

Dashboard devices → board terminals

Device	Output	Ready in	Fault in	Gauge	Valve	Serial
MechanicalVacuumPump	A0 → K1 · J11	B0 → DI1 · J31-1	DI5 · J32-1	AI1 · J41-1/2	V1 · J21-1/2	–
TurboVacuumPump	A1 → K2 · J12	B1 → DI2 · J31-2	DI6 · J32-2	AI2 · J41-3/4	V2 · J21-3/4	RS-485 addr 1
UHVacuumPump_1	A2 → K3 · J13	B2 → DI3 · J31-3	DI7 · J32-3	AI3 · J41-5/6	V4 · J22-3/4	RS-485 addr 2
UHVacuumPump_2	A3 → K4 · J14	B3 → DI4 · J31-4	DI8 · J32-4	AI4 · J41-7/8	V5 · J23-1/2	RS-485 addr 3
HighVoltageTransformer	A4 → K5 · J15	–	–	–	–	–

K1 runs from the E-stop rail; K2–K8 and V1–V8 run from the watchdog rail, which drops about 100 ms after the heartbeat on GPIO19 stops.

Sources

PiShop.us current-boards page (Pi 4B 1/2/3/4/8 GB: $35, $67.50, $83.75, $100, $165; Pi 5 1/2/4/8/16 GB: $45, $77.50, $110, $175, $305; 1 GB and Pi 5 2 GB shown as notify-me), Pi 4 and Pi 5 power-supply pages ($8.80 / $12.95), M.2 HAT+ ($12, in stock) and M.2 HAT+ SSD kit 256 GB ($99) product pages, all 15 Oct 2026. The Register, 2 Oct 2026: 2 GB Pi 4 and Pi 5 raised by $12.50 on memory costs. Board parts: Operations branch, fab/RPi5VacuumIO-bom.csv. Re-check prices before ordering; Pi prices changed several times in 2026.